Question Bank:

CASE1:

SOURCE:
https://www.acs.org.au/memberships/professional-ethics-conduct-and-complaints.html

Ethics Case Study

Synthetic Voice for AI System

Acknowledgement

This sample case study is derived from actual case that is documented in the 2019 CSIRO
Discussion Paper “Artificial Intelligence: Australia’s Ethics Framework”.

Context

Alan and Brad have implemented an AI-based Home Assistant system that responds to voice
commands. They have chosen a female voice to use, based on research that shows that more
people respond to and trust a female voice.

Dilemma

Their own research shows that an overwhelming majority of the users of this system like this voice.
However, this decision comes under much criticism from a vocal lobby group that claims that this
reinforces the subservient female stereotype.

Options

What action can Alan and Brad take?
1. Ignore the criticism as it reflects only a minority.
2. Invite the lobby group to engage in a public debate about the merits or otherwise of this
female voice.
3. Modify the voice to make it more neutral (i.e., of indeterminate sex).
4. Create male and female versions and decide which to use based on user feedback.
5. Create male and female versions and give users the choice of which to use.
6. Something else? Which is…

Ethical Decision

Based on the ethical considerations and practical aspects, option 5 - creating male and female versions and giving users the choice appears to be the most balanced and ethical approach. This addresses user preferences and avoids imposing a single voice type while acknowledging the concerns about gender stereotypes. This option also demonstrates that technology industry is able to adapt its systems.

CASE2:

SOURCE:
https://www.acs.org.au/memberships/professional-ethics-conduct-and-complaints.html

Acknowledgement

This sample case study is derived from actual case that is documented in the 2019 CSIRO
Discussion Paper “Artificial Intelligence: Australia’s Ethics Framework”.

Context

An analysis of judicial parole hearings in a particular jurisdiction revealed there was a distinct
pattern correlating severity of sentences with certain times of the day (i.e. more lenient early in the
morning or straight after lunch, more severe late in the morning or late in the afternoon).

Dilemma

A machine learning (ML) system was trained to analyse all the factors that a judge would consider
along with the judgement in 1,000 cases; it was configured to compensate for those cases
determined late in the morning or late in the afternoon, which had exhibited greater severity of
sentence. The resulting ML system produced much less biased outcomes (sentences), eliminating
the “judge fatigue” pattern.
Unfortunately, because it was a deep learning system, its decision-making logic was opaque, and
therefore it was not possible for the system to provide evidence as to why certain decisions were
proposed.
How would you help the officers in this jurisdiction to decide whether to implement this system as a
replacement for human judges?

Options

The officers of jurisdiction have following options for adopting this system:

1. No, do not implement it until its explicability is improved.
2. No, but it can be used to advise judges in a testing environment to ascertain that no biases
exist within the model and train the system for better results.
3. Yes, but not until it has been trained on many more cases.
4. Yes, but a judge should be able to overrule it.
5. Yes, but the appellant can have the right to appeal to a human judge.
6. Yes, its accuracy is clearly much better than human judges.
7. Something else, which is…

Ethical Decision
Using this system without human judges can lead to incorrect decisions as the system algorithm is not explicable. so here our best option is #1. Option #2 can also be considered valid but only if system is used in a controlled environment under human judges' supervision to notice patterns, understand the system's reasoning and ensure accountability.

CASE3:

SOURCE:
https://www.acs.org.au/memberships/professional-ethics-conduct-and-complaints.html

Acknowledgement
This sample case study is derived from actual case that is documented in the 2019 CSIRO Discussion Paper "Artificial Intelligence: Australia's Ethics Framework".

Context
Rachel, an analyst at a big-data consultancy firm, is conducting research on social media activity.
She gathers posts from the past two years across 10,000 users, ensuring their identities remain
hidden in any published results. To process over 200,000 posts, she employs a machine learning
(ML) system.

Dilemma
In a separate initiative through a data sharing scheme, Rachel acquires 100 million anonymised
posts from 5 million users over two years. She later discovers that the usernames in reposts weren't removed, which compromises the anonymity. Her data science tools and ML system can
identify many usernames in this supposedly anonymised data. This realisation shows that the

additional data greatly enhances the value of her research.

Options
Rachel has several options to choose from. She can:
1. Scrub the data to remove usernames and then run her analysis on properly anonymised data.
2. Immediately discard the identifiable data, advise the supplying agency, and request the data again.
3. Research the supplying agency's data management policies, and if they're in breach of their policy, show them the identifiable data, request that it be replaced with anonymised data, and point out the potential for reputational damage arising from their data management practices.
4. Run the analysis just to see what it would produce, but then discard the identifiable data.
5. Run the analysis and produce and publish the results (not to mention the data breach).

Ethical Decision
The best approach for Rachel would be option 2 (immediately discard the identifiable data) or option 3 (research the agency's data management policies and contact them if required). Doing so ensures she acts with integrity and honesty. Here's why the other options are inadequate: Option 1 – This may be acceptable, but she does have a responsibility to the supplier of those data to ensure that they know and can take whatever corrective action may be required, including removing any intermediate copies of those non-anonymised data. Option 5 – Unethical and not permissible. Option 4 – Also, unethical and not permissible, no matter how tempting and discreet she may be about it.

CASE 4:

Source: https://raeng.org.uk/media/batgo4ye/engineering_ethics_in_practice_shorter.pdf

Dilemma
You have undertaken some consultancy work with a foreign company, under a scheme whereby half of your fee comes from the central government. However, the client company informs you after the work has been done that they are in financial difficulties, and that the only way you will be paid in full is if you falsify the invoice document so that the government pays 100% of your fee. You are also told that this is standard practice, and happens with the cooperation of the administrators of the government fund

What should you do?
1. You could agree to the process as described by the client company. It is important that Sudobuild get fair remittance for the work they have undertaken, and administrators of the government fund have approved the practice of doubling the invoice.

2. You could refuse to participate in the practice, and accept whatever funds that the client company has available. You do not want to engage in corruption, but you do not want to sever your relationship with this company and others in the region.

3. You could refuse to double your invoice, and take the client company to court to recover your fee. It is important to take a stand against corruption, and to ensure that companies face up to their financial obligations

Summary
Several issues need to be balanced in this situation: the reputation of your company, the need to be paid for your work and how to deal with corrupt practices. Going along with the practice may appear to be the easiest option, but it is illegal and would constitute “a deceptive act” and a “corrupt practice”, both of which are explicitly prohibited by the code of conduct. Agreeing to take a smaller amount of money, on the other hand, may result in your company being seen as lacking business acumen, and you may risk other clients trying to do the same with you, resulting in your company not being paid for their work. Equally, while court proceedings may expose the corrupt practice of your client and offer a way of recovering your money, you run the risk of alienating many potential clients in the area.

CASE 5:

source :
https://onlineethics.virginia.edu/collection-detail/Addressing%20Problems%20in%20Research%20Ethics

Description

Joe Engineer encounters a conflict of interest when attempting to support work he performed for a past employer.

Body

Joe Engineer worked for a private engineering company in the field of water rights. The firm was hired by a client to complete a water-rights analysis in which Joe participated. Joe, along with one other employee at the firm, stamped the final document. These types of analyses quantify water rights and provide terms and conditions for future use that must be approved by the local courts. Typically, the court process takes years to complete, and in short, it includes the following steps:

1. Application (proposal)
2. Engineering to support application
3. Objections from the public/other water users
4. Rebuttal of objector's comments
5. Mediation or trial

Joe worked on the project up through step No.2 and resigned from the firm to work for the State. The State is typically an objector in most cases, and it is an objector in this specific analysis.

Joe feels that he can and should support the work he performed and which was included in the stamped report, but he is concerned about the remaining steps in the court process. In his current employment, he has been isolated from the State's case in the matter, and his current position does not include opposing this or other cases.

To what extent is Joe obligated to continue to represent the client? In what ways might this cause conflict with his current employer?

CASE 6:

source :
https://onlineethics.virginia.edu/collection-detail/Addressing%20Problems%20in%20Research%20Ethics

Description

A short case on who should be considered an author.

Body

Andy and John are coauthors who want to submit a manuscript to a journal. Andy thinks the journal co-editor, Candice is prejudiced against him, but that the other co-editor, Donna, is not. Andy thinks John should remove Andy's name from the manuscript and submit it to Donna so that only John is associated with the manuscript. If it is accepted, then Andy proposes to add his name back onto the article's byline.

- Is this an acceptable course of action?

Citation

Description

In 2014, researchers and students at University of Bergen questioned the university's research cooperation with the nation's oil industry, Statoil. The rector asked the Norwegian National Committee for Research Ethics in Science and Technology (NENT) for advice, which it provided in a public report.

Body

NENT provides a variety of resources for Norwegian scientists and engineers and other researchers to address issues of integrity, protection of research participants, and social responsibility. Among them are ethics guidelines, adopted in 2008 and

revised in 2016. This guidance identifies obligations of research to society that include:

- Research has an independent responsibility for the role it plays in social developments.
- Research should be compatible with sustainable development.
- Research has a responsibility to contribute to greater global justice.

The questions the rector posed to NENT were as follows: Is our cooperation with the oil industry ethically questionable? Is petroleum research questionable in light of global warming?

A press release summarizing results of the inquiry reported the NENT conclusion from the Chair as follows: “It is indefensible from a research ethics perspective if petroleum research hinders processes of transition to sustainable energy and thus prevents achievement of UN climate goals which Norway has pledged to uphold.”

Answering these questions adequately requires the collection of factual information and the interpretation of those facts, and recognition that some factual evidence may be lacking while the decision needs to be made. It also requires recognition of the relevant ethics guidelines that can and should be applied to developing these answers. The boundaries of factual information to be collected are affected by the identity of the ethical guidelines.

1. What factual information needs to be collected to answer the questions?
2. Are the ethical guidelines above the right ones to use in this inquiry? Should others be applied?
3. Does this conclusion satisfy the ethical guidance in the three principles above? Is it feasible for researchers to apply these principles to their activities? Is it likely?

The entire NENT statement can be found at: etikkom.no/en/news/news-archive/2015/nents-assessment-of-research-ethics-in-petroleum-research/regarding-the-assessment-of-research-ethics-in-petroleum-research/ . Last updated: Sunday, October 19, 2014; accessed on July 18, 2017.

Considering the full NENT statement, can you identify one or two positions in it that rely on one or another of the three principles above for their justification? Can you identify what principles were affecting what information the committee collected? Did the committee collect the information it needed to answer the questions?

CASE 7:

CASE 8:

CASE 9:

CASE 10: